

LAB 6 REPORT
THE INFLUENCE OF ANGLE OF ATTACK ON LIFT AND DRAG

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1 INTRODUCTION

For aircraft, the wing geometry is one of the biggest factors affecting the lift and drag of the aircraft (Benson, n.d.), and optimizing these values is crucial for developing more efficient aircraft. More efficient aircraft is a necessity for the future of air travel as it makes the aircraft more fuel efficient and easier to make net carbon zero, which is a major goal by the year 2050 (Rutherford, 2024). Our objective is to calculate and plot the lift and drag coefficients of various airfoils (NACA 0012, NACA 2412, NACA 2415, and NACA 4412) at differing angles of attack to see the performance characteristics of each of them.

The coefficient and drag and lift can easily be calculated with varying angles of attack in a subsonic wind tunnel using a multitude of assumptions and equations. One of these assumptions is that we ignore the friction due to the edges of the wind tunnel. We also assume that the air is all the same density and viscosity. Lastly, we ignore the effects of turbulent flow around the stand.

Equation 1 below shows the calculation for the wind speed of the wind tunnel, U_∞ , using the dynamic pressure of the wind tunnel, P , as well as the density of the air, ρ .

$$U_\infty = \frac{2P}{\rho} \quad (\text{Eq. 1})$$

Equation 2 below calculates the change in height of the front of the airfoil compared to the back, h . This is calculated from the known angle of attack of the airfoil in radians, α .

$$h = \frac{1}{2} \tan(\alpha) \quad (\text{Eq. 2})$$

Equations 3 and 4 below are the equations for the lift and drag of the model only. This is found by subtracting the drag/lift of the model and the drag/lift of the stand.

$$D_{\text{model only}} = D_{\text{model}} - D_{\text{stand}} \quad (\text{Eq. 3})$$

$$L_{\text{model only}} = L_{\text{model}} - L_{\text{stand}} \quad (\text{Eq. 4})$$

Equations 5 and 6 below calculate the lift and drag considering the calibration of the wind tunnel. $A_{\text{drag/lift}}$ is the slope of the lift/drag calibration graph (graphs shown in the Appendix A). $B_{\text{drag/lift}}$ is the y-intercept of that same lift/drag calibration graph. $D/L_{\text{model only}}$ is the drag or lift of the model.

$$D_{calibration} = A_{drag}D_{model\ only} + B_{drag} \quad (\text{Eq. 5})$$

$$L_{calibration} = A_{lift}L_{model\ only} + B_{lift} \quad (\text{Eq. 6})$$

Equation 7 and 8 below solves for the coefficient of drag of the airfoil(c_d) and the coefficient of lift of the airfoil(c_l) respectively. ρ is the air density in the wind tunnel. U_∞ is the wind speed of the wind tunnel. S is the reference area of the airfoil. Equation 9 below shows the equation for the reference a area, where b is the span and D is the chord length of the airfoil.

$$c_d = \frac{2D_{calibration}}{\rho U_\infty^2 S} \quad (\text{Eq. 7})$$

$$c_l = \frac{2L_{calibration}}{\rho U_\infty^2 S} \quad (\text{Eq. 8})$$

$$S = bD \quad (\text{Eq. 9})$$

The Reynolds number is also crucial, and equation 10 below shows the calculation for it where ρ is the air density, D is the chord length of the airfoil, μ is the viscosity of the air in the wind tunnel, and U_∞ is the wind speed of the wind tunnel.

$$Re = \frac{\rho U_\infty D}{\mu} \quad (\text{Eq. 10})$$

2 OBJECTIVE

The objective of this experiment is to measure the lift and drag forces acting on four different NACA airfoils (NACA 0012, NACA 2412, NACA 2415, and NACA 4412) over a range of angles of attack from 0° to 15° using a low-speed wind tunnel operating at a constant velocity. By recording and analyzing the lift and drag forces at approximately 2.5° increments of angle of attack, we aim to calculate the lift and drag coefficients for each airfoil and compare the experimental results to published NACA data. This will allow us to evaluate the aerodynamic performance characteristics of each airfoil and assess how changes in airfoil geometry affect lift and drag, all within the scheduled laboratory time frame.

3 METHOD

3.1 *Experiment Setup and Test Model(s)*

Lab Location: ARMS B098, Low Speed Wind Tunnel

The low-speed wind tunnel was used to simulate continuous, uniform, steady flow with a constant free stream velocity that surrounds the airfoils. A force balance was used along with the LabView software to measure the lift and drag experienced by the airfoils as a function of angle of attack. A stand was placed on top of the force balance to hold the test models in place while in the wind tunnel during the experiment. The airfoils tested in this experimentation were the NACA 0012, NACA 2412, NACA 2415, and NACA 4412. A caliper was used to measure the chord length and wingspan of the airfoils in order to calculate the reference area. A torpedo level, along with an inclinometer, ensured the airfoils were placed in the wind tunnel with an initial angle of attack of 0° .



Figure 4: NACA 0012 Airfoil

Test models used in this experiment were almost identical in dimension and material except for one airfoil, which was made of metal. NACA 0012, NACA 2412, and NACA 4412 were 3-D printed using ABS material, while NACA 2415 is composed of metal. Prior to the test of the airfoils, the stand was placed in the wind tunnel and tested under the same free stream velocity as that of the airfoils to calculate the lift and drag produced by the stand. The airfoils were then fastened to the stand in two spots along the airfoils' chord using screws, ensuring that only the angle of attack is changing throughout the experiment.

3.2 Instrumentation

Low-speed wind tunnel: Provided airflow to simulate a continuous, steady, uniform flow in which the test models were tested. For all test models, the wind tunnel was set to provide airflow with a frequency of 20 Hz.

Force balance: Measured lift and drag forces on the stand and test models with a precision of 0.001 lb. The force balance and test models were connected to the stand, and all the forces exerted on the airfoil were passed to the force balance through the stand.

Calipers: Used to measure the chord length of the test models. The calipers have a measuring range between 0 and 15 centimeters, with a precision of 0.01 centimeters.

Ruler: Used to measure the wingspan of the test models. The ruler's measuring range is between 0 and 30 centimeters, with a precision of 0.1 centimeters.

Torpedo level: Used to ensure a surface is horizontal to the ground by observing if the bubble is in the center of the liquid window. Used to ensure the test model is placed at a 0° angle of attack in the wind tunnel initially.

Inclinometer: Used to measure the angle between a certain surface and ground level, ranging from 0° to 90° . It is clamped at the leading edge and trailing edge of the airfoil models to ensure the angle of attack of the airfoil was 0° when initially placed in the wind tunnel.

3.3 Experimental Conditions

The test conditions within the low speed wind tunnel were close to ideal, although the tunnel's flow wasn't perfectly smooth because the diffuser at the inlet was slightly damaged. The flow speed was relatively low at 13.34 m/s for all the airfoils, same thing for the Reynolds number which was also relatively small at 108491.11. Aside from the shortcomings of the airflow, the tunnel was not perfectly sealed, although an effort was made to ensure any ports were sealed.

Table 3.3.1

Airfoil	Flow Speed (m/s)	Reynolds Number
NACA 0012, NACA 2412, NACA 2415, NACA 4412	13.34	108491.11

3.4 Procedures

Before beginning measurements, the lift and drag sensors on the force balance were calibrated. To calibrate lift, known weights ranging from 0.5 lb to 5 lb were placed on top of the force balance,

and the corresponding sensor readings were recorded using the WindTunnelBalance VI in LabVIEW. A linear calibration curve of "Actual Weight" versus "Measured Weight" was generated for lift calibration. Similarly, to calibrate drag, the same weights were placed on the pulley mechanism attached to the drag transducer, and a separate drag calibration curve was produced.

After completing the calibration, one of the 3D-printed NACA airfoils (NACA 0012, NACA 2412, NACA 2415, or NACA 4412) was securely mounted onto the force balance stand, ensuring the tunnel was sealed properly. The LabVIEW program was restarted to zero out the forces before the wind tunnel was turned on. The wind tunnel was then turned on and set to a fan speed of 20 Hz, corresponding to a free-stream velocity measured with a digital manometer. This velocity remained constant throughout all tests.

The angle of attack was controlled using the micrometer attached to the adjustment stand. Before testing, a pre-determined table correlating micrometer height changes to angle of attack increments (in degrees) was referenced to calculate the required adjustment for each test point. The micrometer was adjusted accordingly to achieve approximately 2.5° increments in angle of attack, starting from 0° and increasing up to 15° , and the corresponding lift and drag forces were measured at each setting. After the wind tunnel flow stabilized at each new angle, the lift and drag value was recorded alongside the current micrometer reading.

This process was repeated across the full range of angles for each of the four NACA airfoils. After all measurements were completed, the airfoil was removed, and the next airfoil was mounted, following the same procedure. After testing all four airfoils, the wind tunnel was turned off, and the lab area was cleaned and restored to its original state.

4 RESULT

The set-up included finding constants that would be useful throughout the duration of the lab. Two of these were the air density and viscosity in the wind tunnel as well as finding the dynamic pressure to calculate the wind speed, which are all referenced in table 4.1 below. Also shown in table 4.1 is the lift and drag of the stand that the model is set on.

Table 4.11

Air Density (kg/m^3)	1.225
Viscosity (Ns/m^2)	0.00001789
Wind Tunnel Frequency (Hz)	20
Dynamic Pressure (kPa)	0.109
Dynamic Pressure (Pa)	109
Wind Speed (m/s)	13.34013432
Lift of Stand (lb)	0.03
Drag of Stand (lb)	0.045

Also calculated is the distance needed for the front of the airfoil to be lifted to obtain the correct angle of attack (h). This is calculated using equation 2 in the introduction and is displayed in Table 4.2 below.

Table 4.22

Angle of Attack (deg)	H value
0	0
2.5	0.021830471
5	0.043744332
7.5	0.065826249
10	0.08816349
12.5	0.110847331
15	0.133974596

4.1 NACA 0012

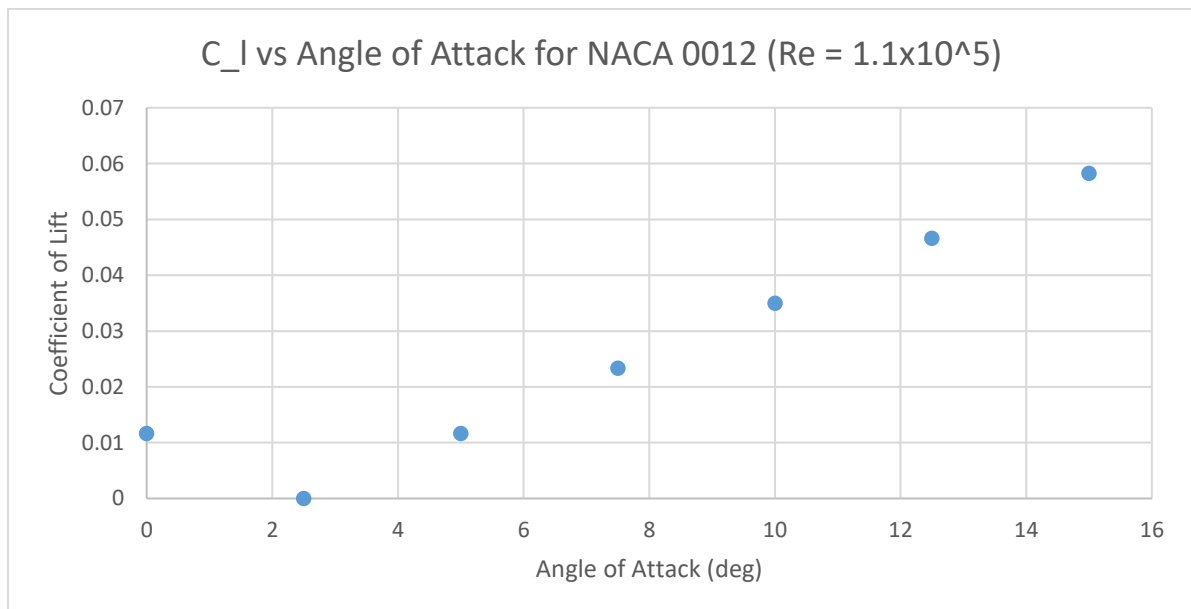


Figure 4.1.1: Coefficient of Lift vs Angle of Attack for NACA 0012

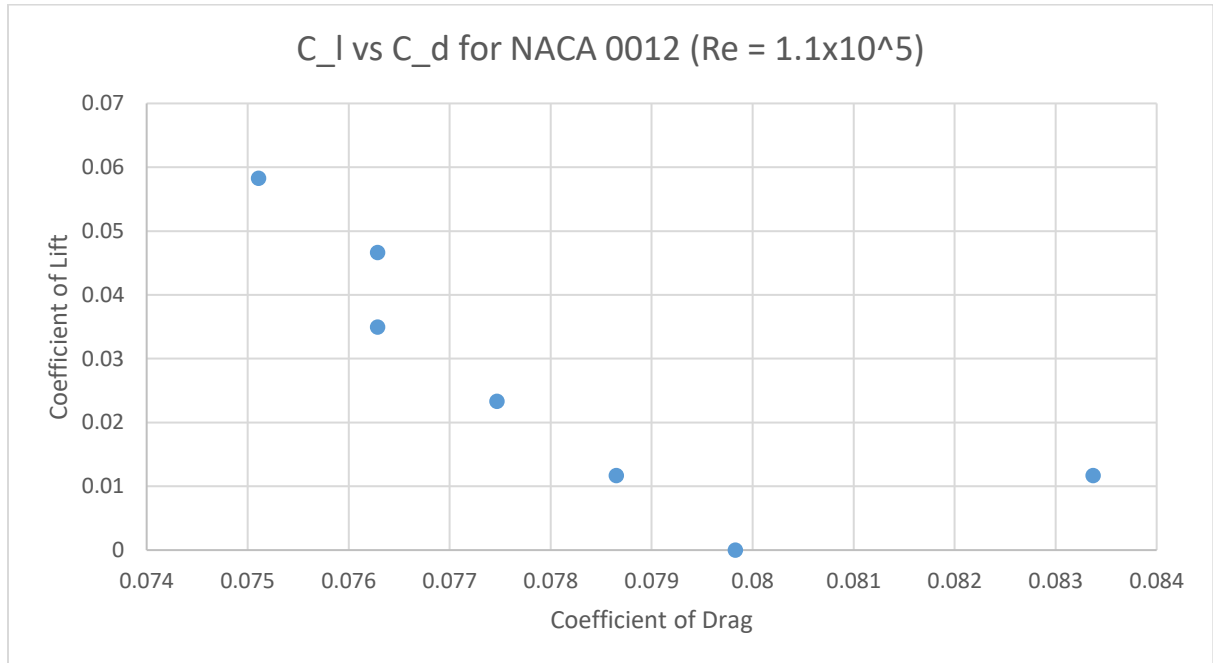


Figure 4.1.2: Coefficient of Lift vs Coefficient of Drag for NACA 0012

The above graphs in figures 4.1.1 and 4.1.2 show the coefficient of lift versus angle of attack and coefficient of drag. The first graph shows that the angle of attack increases with the coefficient of lift. The second graph shows that the coefficient of lift decreases with increasing coefficient of drag, excluding the outlier at a coefficient of drag of 0.083.

The Reynolds number was calculated to be approximately 108491 using equation 10 in the introduction, while the span and chord of the airfoil were found to be 0.119 meters and 0.297 meters respectively.

Table 4.1.1

Angle of Attack (deg)	Drag of the Model (lb)	Drag of the Model Only (lb)	Drag from Calibration (lb)	Drag in Newtons	Cd
0	0.107	0.062	0.0721082	0.320753137	0.083371308
2.5	0.104	0.059	0.0690449	0.307126905	0.079829529
5	0.103	0.058	0.0680238	0.302584828	0.078648936
7.5	0.102	0.057	0.0670027	0.29804275	0.077468343
10	0.101	0.056	0.0659816	0.293500673	0.07628775
12.5	0.101	0.056	0.0659816	0.293500673	0.07628775
15	0.1	0.055	0.0649605	0.288958595	0.075107157

Table 4.1.2

Angle of Attack (deg)	Lift of the Model (lb)	Lift of Model Only (lb)	Lift from Calibration (lb)	Lift in Newtons	Cl
0	0.02	0.01	0.010075	0.044815817	0.011648688
2.5	0.03	0	0	0	0
5	0.04	0.01	0.010075	0.044815817	0.011648688
7.5	0.05	0.02	0.02015	0.089631633	0.023297376
10	0.06	0.03	0.030225	0.13444745	0.034946064
12.5	0.07	0.04	0.0403	0.179263266	0.046594752
15	0.08	0.05	0.050375	0.224079083	0.058243441

The tables above (4.1.1 and 4.1.2) show the calculated coefficient of lift and coefficient of drag for each angle of attack. They also show the retrieved data such as the lift and drag of the model, as well as the calculated values of the lift/drag of the model only and the lift/drag of the calibration.

4.2 NACA 2412

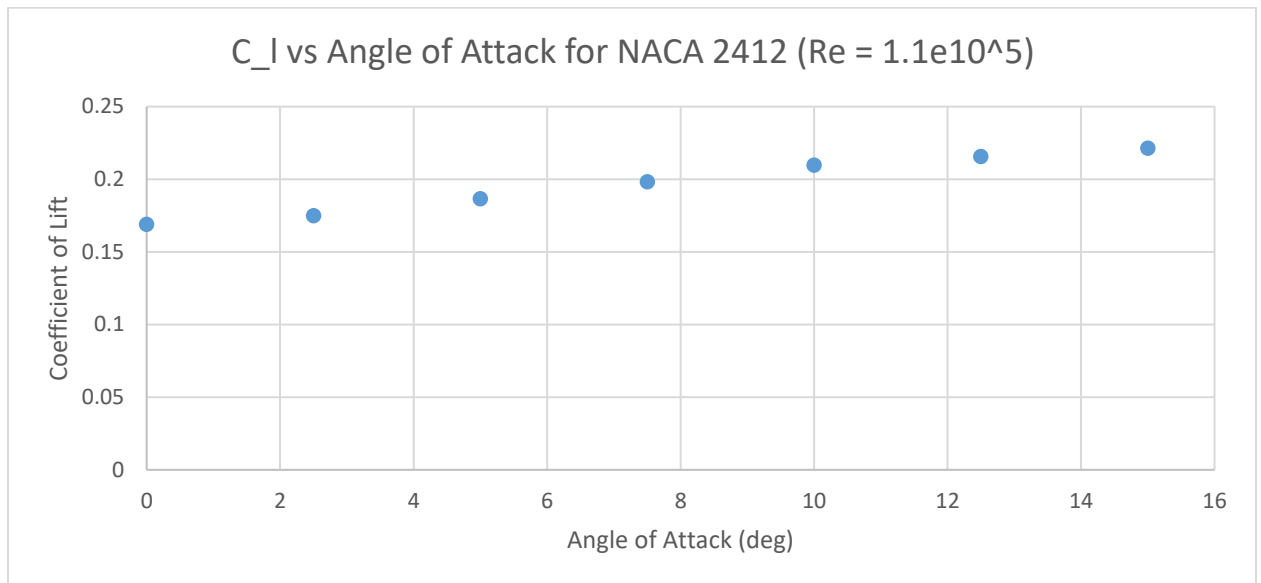


Figure 4.2.1: Coefficient of Lift vs Angle of Attack for NACA 2412

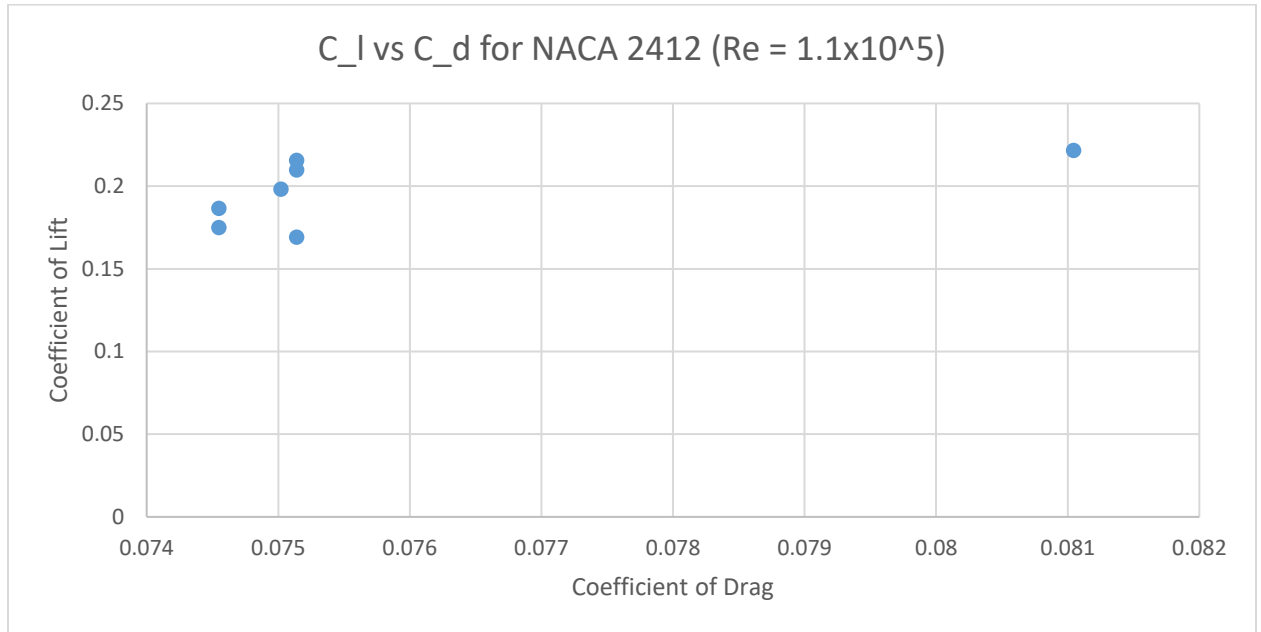


Figure 4.2.2: Coefficient of Lift vs Coefficient of Drag for NACA 2412

The above graphs in figures 4.2.1 and 4.2.2 show the coefficient of lift versus angle of attack and coefficient of drag. The first graph shows that the angle of attack increases the coefficient of lift. The second graph has a lower R^2 value, but generally shows a trend of increasing coefficient of lift with increasing coefficient of drag.

The Reynolds number was calculated to be approximately 108491 using equation 10 in the introduction, while the span and chord of the airfoil were found to be 0.119 meters and 0.297 meters respectively.

Table 4.2.1

Angle of Attack (deg)	Drag of the Model (lb)	Drag of the Model Only (lb)	Drag from Calibration (lb)	Drag in Newtons	Cd
0	0.1	0.055	0.0649605	0.288958595	0.075107157
2.5	0.0995	0.0545	0.06444995	0.286687557	0.07451686
5	0.0995	0.0545	0.06444995	0.286687557	0.07451686
7.5	0.0999	0.0549	0.06485839	0.288504388	0.074989098
10	0.1	0.055	0.0649605	0.288958595	0.075107157
12.5	0.1	0.055	0.0649605	0.288958595	0.075107157
15	0.105	0.06	0.070066	0.311668983	0.081010122

Table 4.2.2

Angle of Attack (deg)	Lift of the Model (lb)	Lift of Model Only (lb)	Lift from Calibration (lb)	Lift in Newtons	Cl
0	0.175	0.145	0.1460875	0.649829339	0.168905978
2.5	0.18	0.15	0.151125	0.672237248	0.174730322
5	0.19	0.16	0.1612	0.717053064	0.18637901
7.5	0.2	0.17	0.171275	0.761868881	0.198027698
10	0.21	0.18	0.18135	0.806684697	0.209676386
12.5	0.215	0.185	0.1863875	0.829092605	0.21550073
15	0.22	0.19	0.191425	0.851500514	0.221325074

The tables above (4.2.1 and 4.2.2) show the calculated coefficient of lift and coefficient of drag for each angle of attack. They also show the retrieved data such as the lift and drag of the model, as well as the calculated values of the lift/drag of the model only and the lift/drag of the calibration.

4.3 NACA 2415

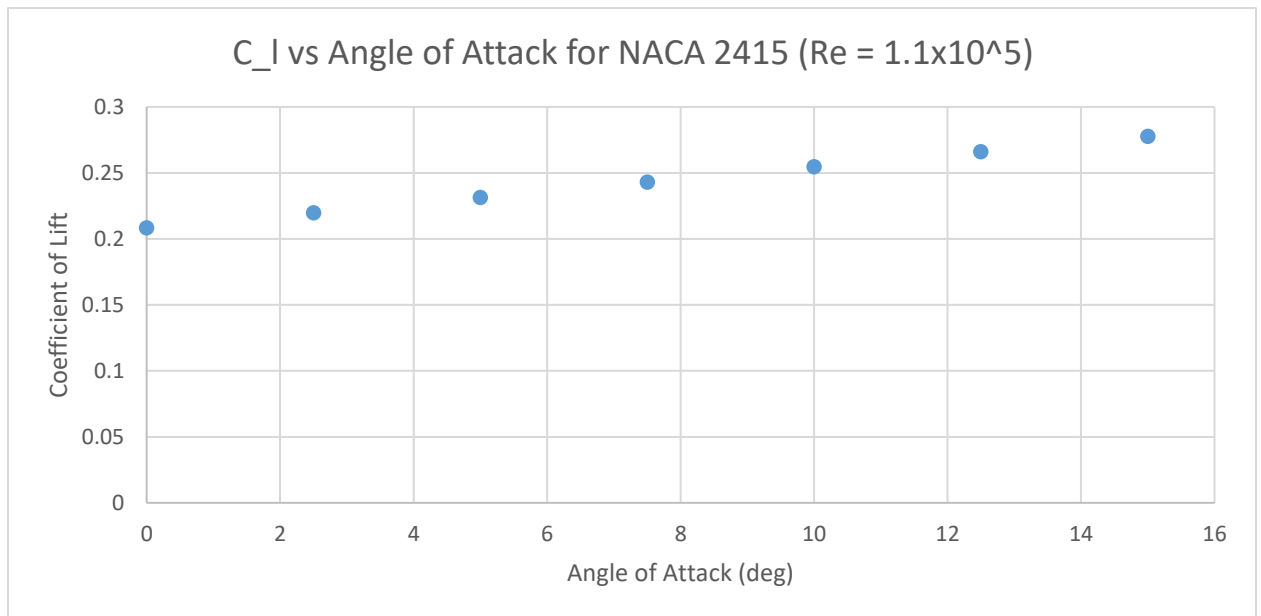


Figure 4.3.1: Coefficient of Lift vs Angle of Attack for NACA 2415

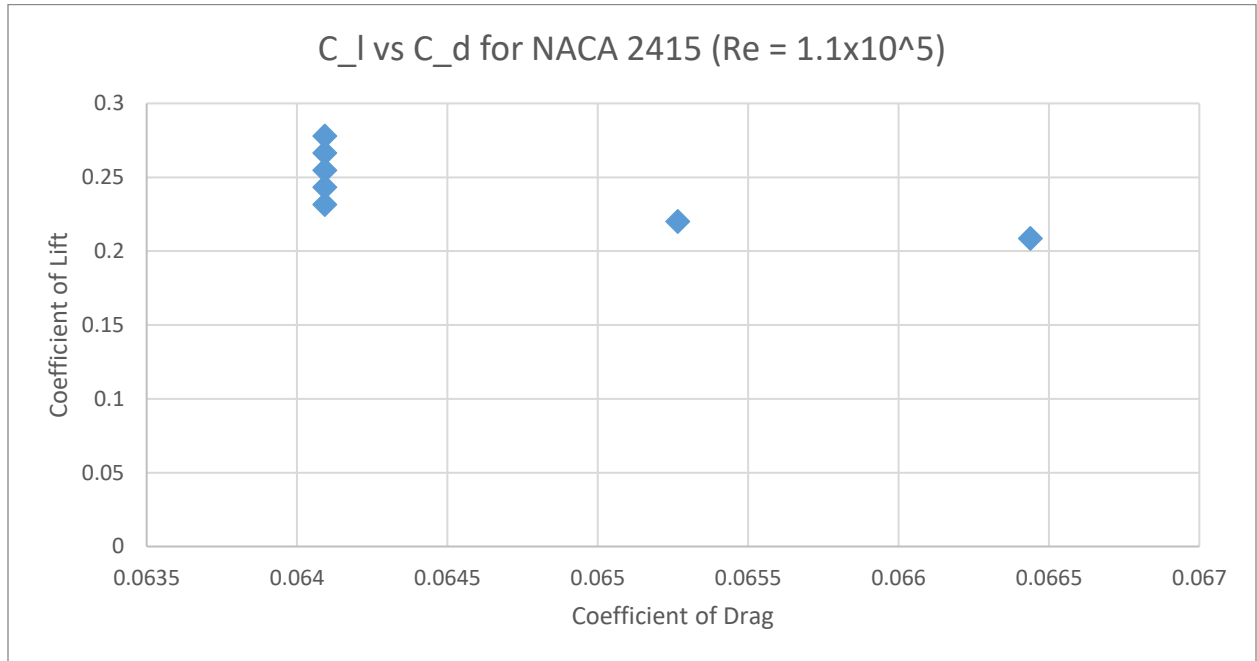


Figure 4.3.2: Coefficient of Lift vs Coefficient of Drag for NACA 2415

The above graphs in figures 4.3.1 and 4.3.2 show the coefficient of lift versus angle of attack and coefficient of drag. The first graph shows that the angle of attack increases the coefficient of lift. The second graph shows that the coefficient of drag stays constant, except for at a coefficient of lift of 0.208 and 0.220. These two points make a trend of decreasing coefficient of lift with increasing coefficient of drag, but due to the low Reynolds Number, this is not a good estimate.

The Reynolds number was calculated to be approximately 108491 using equation 10 in the introduction, while the span and chord of the airfoil were found to be 0.121 meters and 0.295 meters respectively.

Table 4.3.1

Angle of Attack (deg)	Drag of the Model (lb)	Drag of the Model Only (lb)	Drag from Calibration (lb)	Drag in Newtons	Cd
0	0.093	0.048	0.0578128	0.257164053	0.066438853
2.5	0.092	0.047	0.0567917	0.252621976	0.065265398
5	0.091	0.046	0.0557706	0.248079898	0.064091943
7.5	0.091	0.046	0.0557706	0.248079898	0.064091943
10	0.091	0.046	0.0557706	0.248079898	0.064091943
12.5	0.091	0.046	0.0557706	0.248079898	0.064091943
15	0.091	0.046	0.0557706	0.248079898	0.064091943

Table 4.3.2

Angle of Attack (deg)	Lift of the Model (lb)	Lift of Model Only (lb)	Lift from Calibration (lb)	Lift in Newtons	Cl
0	0.21	0.18	0.18135	0.806684697	0.208408622
2.5	0.22	0.19	0.191425	0.851500514	0.219986878
5	0.23	0.2	0.2015	0.89631633	0.231565135
7.5	0.24	0.21	0.211575	0.941132147	0.243143392
10	0.25	0.22	0.22165	0.985947963	0.254721649
12.5	0.26	0.23	0.231725	1.03076378	0.266299905
15	0.27	0.24	0.2418	1.075579596	0.277878162

The tables above (4.3.1 and 4.3.2) show the calculated coefficient of lift and coefficient of drag for each angle of attack. They also show the retrieved data such as the lift and drag of the model, as well as the calculated values of the lift/drag of the model only and the lift/drag of the calibration.

4.4 NACA 4412

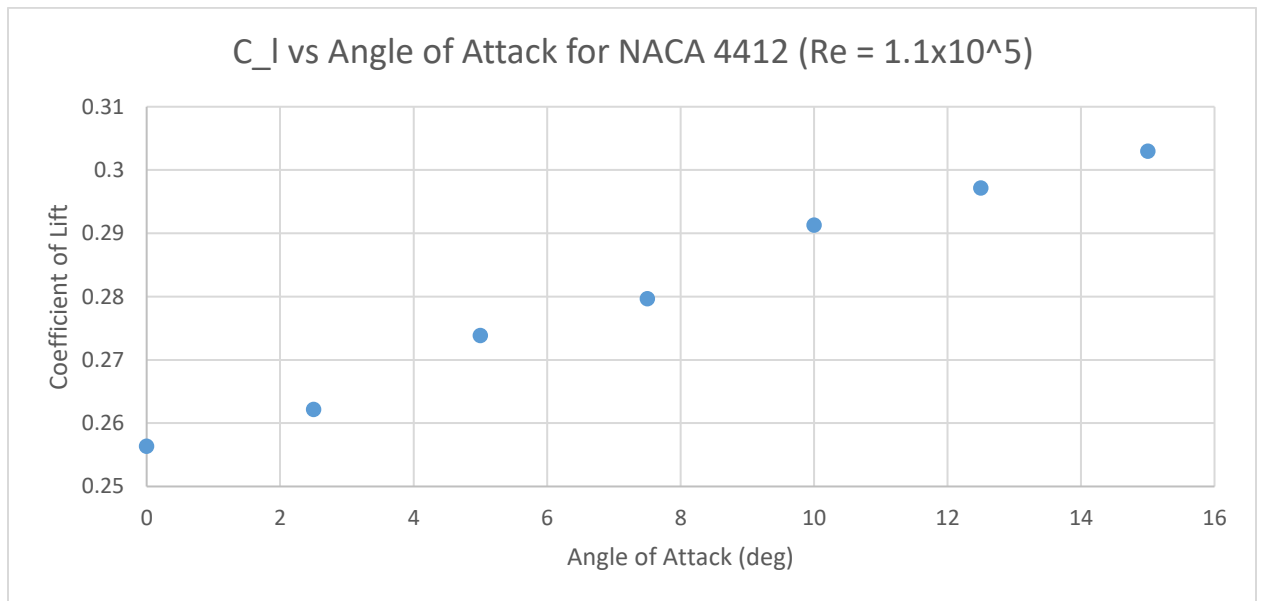


Figure 4.4.1: Coefficient of Lift vs Angle of Attack for NACA 4412

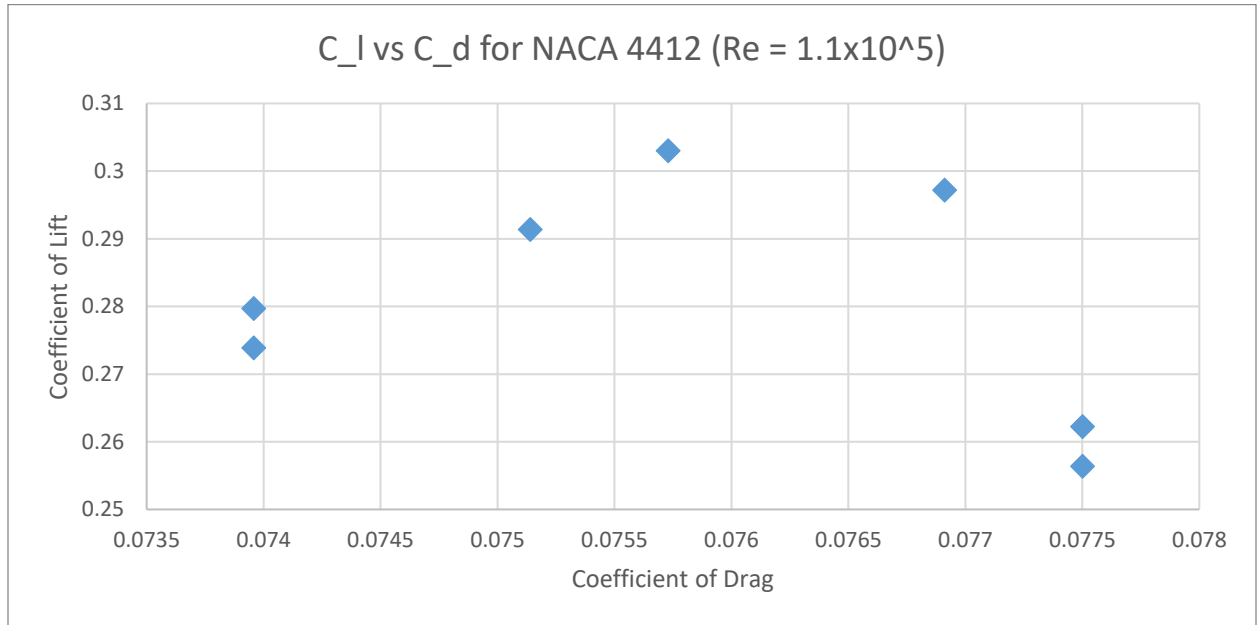


Figure 4.4.2: Coefficient of Lift vs Coefficient of Drag for NACA 4412

The above graphs in figures 4.4.1 and 4.4.2 show the coefficient of lift versus angle of attack and coefficient of drag. The first graph shows that the angle of attack increases the coefficient of lift. The second graph has a lower R^2 value but generally shows an increasing coefficient of lift with increasing coefficient of drag until the maximum coefficient of drag, which has a much lower coefficient of lift.

The Reynolds number was calculated to be approximately 108491 using equation 10 in the introduction, while the span and chord of the airfoil were found to be 0.119 meters and 0.297 meters respectively.

Table 4.2.1

Angle of Attack (deg)	Drag of the Model (lb)	Drag of the Model Only (lb)	Drag from Calibration (lb)	Drag in Newtons	Cd
0	0.102	0.057	0.0670027	0.29804275	0.077501492
2.5	0.102	0.057	0.0670027	0.29804275	0.077501492
5	0.099	0.054	0.0639394	0.284416518	0.073958197
7.5	0.099	0.054	0.0639394	0.284416518	0.073958197
10	0.1	0.055	0.0649605	0.288958595	0.075139295
12.5	0.1015	0.0565	0.06649215	0.295771711	0.076910942
15	0.1005	0.0555	0.06547105	0.291229634	0.075729844

Table 4.2.2

Angle of Attack (deg)	Lift of the Model (lb)	Lift of Model Only (lb)	Lift from Calibration (lb)	Lift in Newtons	Cl
0	0.25	0.22	0.22165	0.985947963	0.256380797
2.5	0.255	0.225	0.2266875	1.008355871	0.262207633
5	0.265	0.235	0.2367625	1.053171688	0.273861306
7.5	0.27	0.24	0.2418	1.075579596	0.279688142
10	0.28	0.25	0.251875	1.120395413	0.291341815
12.5	0.285	0.255	0.2569125	1.142803321	0.297168651
15	0.29	0.26	0.26195	1.165211229	0.302995487

The tables above (4.2.1 and 4.2.2) show the calculated coefficient of lift and coefficient of drag for each angle of attack. They also show the retrieved data such as the lift and drag of the model, as well as the calculated values of the lift/drag of the model only and the lift/drag of the calibration.

One source of error could have been that the airfoil was shaking during the wind tunnel testing, disrupting the flow. This is due to the fact that the 3D modeled airfoils were plastic, so the screws were not fully tightened to prevent breaking the airfoils. Although to a lesser extent there was the same issue with the metal airfoil. The shaking could have caused errors in the lift and drag of the model, as the air would not have been completely laminar.

5 DISCUSSION

The objectives of this lab were met in part by the successful collection of data on lift and drag from the varying airfoils, but also by the failure to define the stall angle for each airfoil. The data recorded created trends similar to the given NACA plots, showing a positive linear slope for the angle of attack vs coefficient of lift. However, this experiment failed to acknowledge the stall angle, omitting data that would have discussed the relationship of angle of attack and the coefficient of lift. An area that was unexpected were the recordings for the coefficient of drag, as the values were slightly lowered as the angle of attack was raised. This phenomenon is likely caused by a low Reynolds number due to a more laminar flow that stuck to the body of the airfoils, causing less separation. Generally, we would see a more coherent relationship with the coefficient of drag once the airfoil hits the stall angle, which would allow lift to decrease significantly due to the flow separation, causing turbulent flow on the top of the wing.

Several limitations were imposed onto this experiment during the testing process. One primary limitation was that the wind tunnel used in this experiment was a low-speed wind tunnel, which did not allow a high Reynolds number to be produced. This can be improved by testing the airfoils in a larger and more professional grade high speed wind tunnel, if the budget allows. Another limitation was the reliability of the angle adjuster stand, as the accuracy of the airfoil being the exact angle of attack dropped due to measuring issues. Again,

with a larger budget, more reliable sensors and perhaps an upgraded angle adjuster stand could improve the accuracy of the data being recorded.

Disregarding equipment issues, the experiment would have been proven to be more successful if more angles of attack beyond the 15-degree point were recorded. By only slightly increasing the angle of attack, the data yielded results that only showed a constant slope for the coefficient of lift and the angle of attack. This group was not able to determine the stall angle for all four airfoils in this experiment, which was one of the primary objectives of this lab.

Overall, the results anticipated lined up with the data recorded during the experiment. The coefficient of lift had a constant slope as the angle of attack increased, and as predetermined, there was a slight increase in the coefficient of lift at zero degrees for the NACA 0012 airfoil. This was likely due to the shape of the airfoil, being more symmetric along the camber line, allowing a small amount of lift. The objectives of this lab were met with somewhat satisfaction and areas of growth, as the data recorded yielded trends that were almost linear, which was the expected trend, but lacked the data for the stall angle. As mentioned previously, implementing more angles of attack to higher degrees can produce data for the stall angle, which then we would be able to analyze the data to its anticipated results.

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APPENDIX

Link to Excel Sheet:

<https://1drv.ms/x/c/5d3e9ffd3bf2bf77/EcGRs5dS7PdAvjxzTfOObCABKDd8k6M7P0nk8l1hCNhJaw?e=Kpt8nW>

APPENDIX A: CALIBRATION OF WIND TUNNEL